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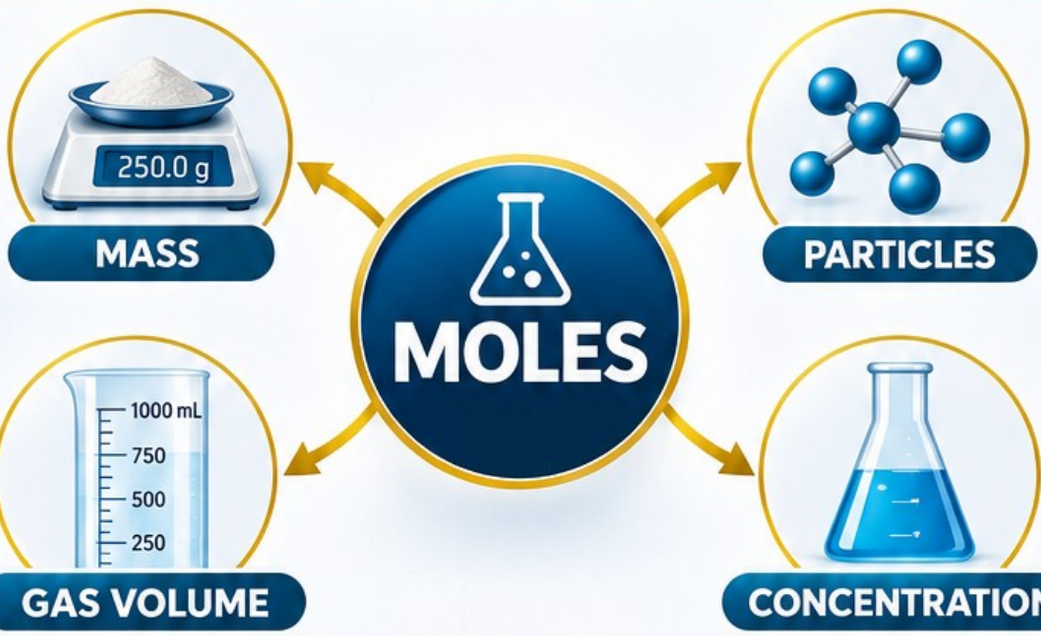
SECOND EDITION • 2027

THINK SMART SERIES

$2H_2 + O_2 \longrightarrow 2H_2O$

THE MOLE CONCEPT CURE

Understand the Mole. Master the Calculations. Conquer the Questions.



A Complete Guide to Chemical Calculations, Stoichiometry and the Mole Concept.

FOR JAMB UTME • WAEC • NECO

BY TECHMED

THE MOLE CONCEPT CURE

Understand the Mole. Master the Calculations. Conquer the Questions.

Think Smart Series

Second Edition, 2027

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Examination questions. Questions labelled [JAMB], [WAEC] or [NECO] are past examination questions whose source has been identified. Questions labelled [TECHMED Practice Question] are original from TECHMED. Every answer in this book has been solved independently; where a commonly circulated answer key is wrong, the book says so.

Accuracy. Every effort has been made to ensure that the content is accurate and current. Students should always confirm examination requirements against the latest official syllabus.

Website: techmedng.com

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About This Book

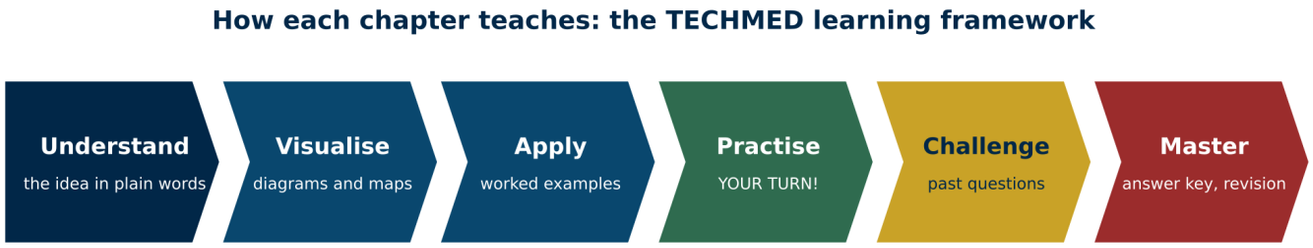
The mole is the single most important calculation idea in secondary-school chemistry, and the one that frightens students most. It appears in formulae, equations, gases, solutions, titrations, electrolysis and organic chemistry, and it is tested in almost every JAMB, WAEC and NECO Chemistry paper.

The Mole Concept Cure removes the confusion by teaching one idea properly: **convert what you are given into moles, use the ratio, and convert into what is required**. Every chapter applies that idea to a new area, using the same five-step method, so that by the end of the book an unfamiliar question feels familiar.

The book is written for SSCE (WAEC and NECO) and UTME candidates, with UTME objective questions given priority. It is also suitable for any student who wants a firm foundation in chemical calculations before university.

> **KEY CONCEPT: The TECHMED philosophy: remove concept chaos**
>
> You should never have to fight through confusing explanations or disconnected formulae before you understand an idea. Each concept is introduced simply, built up step by step, shown in a diagram, applied in a worked example, and then practised.

How to Use This Book



Understand → Visualise → Apply → Practise → Challenge → Master

Figure F.1 — The TECHMED learning framework

1. **Work through Chapters 1–7 in order.** Each builds on the one before; Chapters 1–3 are the foundation for everything else.
2. **Study every worked example** ("LET'S SOLVE TOGETHER") with a pen. Cover the solution and try each step before reading it.
3. **Attempt every YOUR TURN! question** before checking the answers at the end of the chapter.
4. **Do the chapter practice questions** under timed conditions, then mark them using the answer key and explanations.
5. **Use Chapter 8** to train pattern recognition, **Chapter 9** for timed question-bank practice, and **Chapter 10** for your final revision plan.
6. **Keep an error log** (Chapter 10). A wrong answer is useful only when you know why it was wrong.

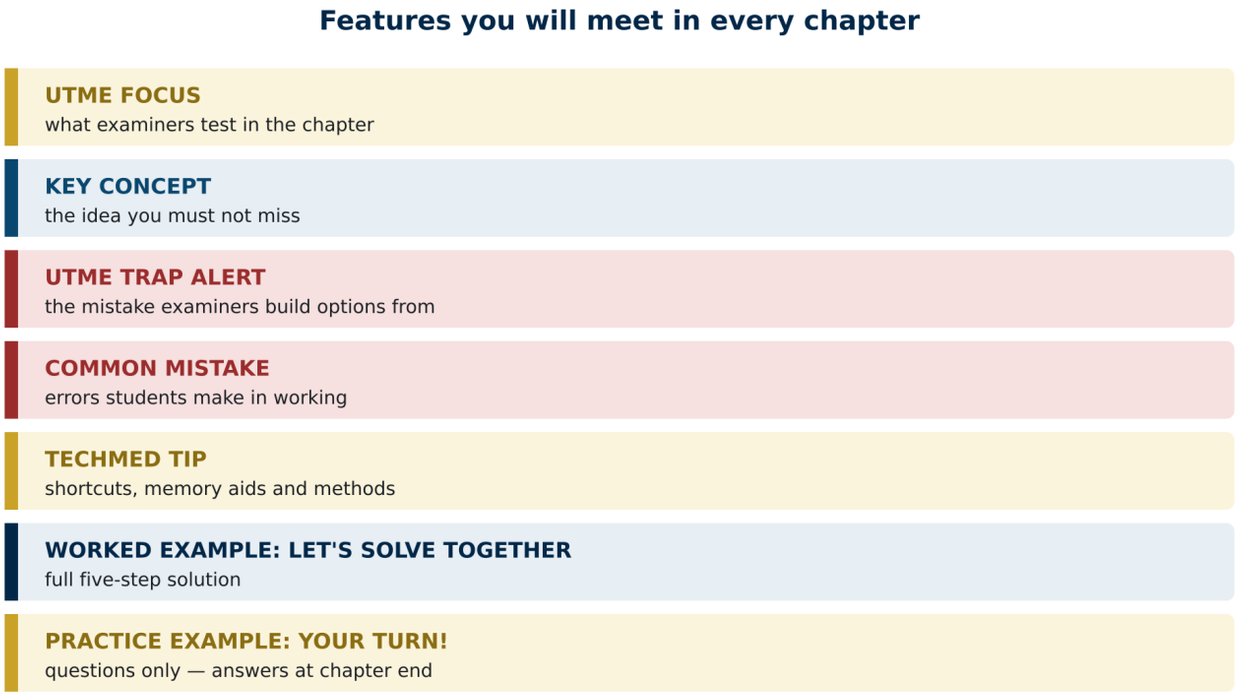
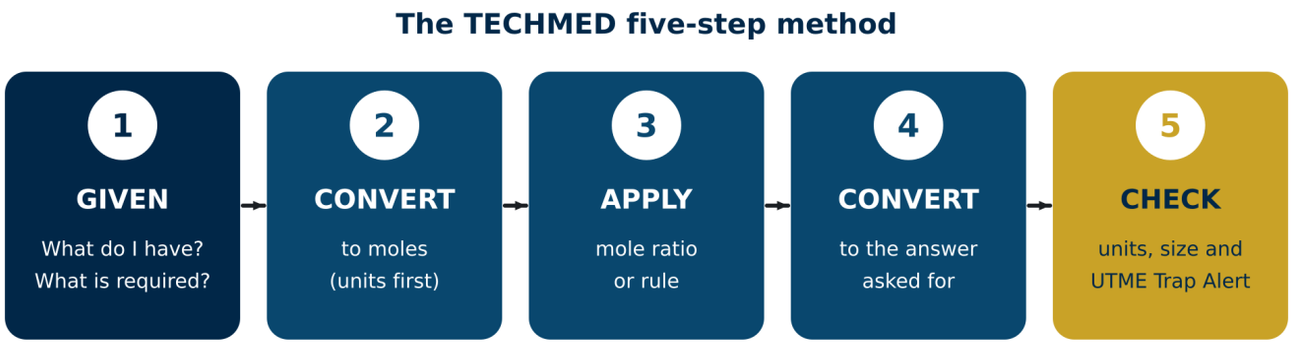


Figure F.2 — Features in every chapter

Question label	Meaning
[JAMB] [WAEC] [NECO]	Past examination question; source identified
[TECHMED Practice Question]	Original question written for this book

The TECHMED Five-Step Method



Used in every worked example in this book. Make it automatic.

Figure F.3 — The five steps used in every worked example

Step	What you do
1. Given	Write down what you have (with units) and exactly what is required.
2. Convert to moles	Convert units first (cm^3 → dm^3, kg → g, min → s), then turn the given quantity into moles.
3. Apply the ratio or rule	Use the mole ratio from the balanced equation, the formula, or the electrode equation.
4. Convert to the answer	Turn moles into the mass, volume, concentration, particles or charge asked for.
5. Check (UTME Trap Alert)	Check units, size and the trap the examiner has set.

Symbols, Units and Constants

Symbol	Quantity	Unit
n	amount of substance	mol
m	mass	g
M	molar mass	$g\ mol^{-1}$
A_r, M_r	relative atomic / molecular mass	no unit
N	number of particles	no unit
N_a	Avogadro constant = 6.02×10^{23}	mol^{-1}
V	volume	cm^3 or dm^3
V_m	molar volume = 22.4 (s.t.p.) or 24 (r.t.p.)	$dm^3\ mol^{-1}$
C	molar concentration	$mol\ dm^{-3}$
P	pressure	atm, mmHg, Pa
T	absolute temperature (°C + 273)	K
R	gas constant = 8.314	$J\ mol^{-1}\ K^{-1}$
Q	quantity of electricity	C
I	current	A
t	time	s
F	Faraday constant = 96 500	$C\ mol^{-1}$
z	electrons per ion in the electrode equation	no unit
V.D.	vapour density (M = 2 × V.D.)	no unit

s.t.p. = standard temperature and pressure (0 °C, 1 atm); r.t.p. = room temperature and pressure. Always use the values given in a question.

Chapter 1 — Understanding the Mole Concept

The foundation of every mole-concept calculation.

> **UTME FOCUS: What UTME tests from this chapter**
>
> Expect direct, quick calculations: **moles from mass**, **mass from moles**, **number of atoms or molecules** in a given mass, **molar mass** of a compound, and **relative atomic mass from isotopes**. The commonest trap is counting **molecules** when the question asks for **atoms** (or the reverse). Every one of these is practised in this chapter.

What You Will Learn

By the end of this chapter, you should be able to:

1. Explain why chemists use the mole concept.
2. Define a mole and state the Avogadro constant.
3. Distinguish between atoms, molecules, ions and formula units.
4. Explain relative atomic mass and calculate it from isotopic abundances.
5. Calculate relative molecular mass, relative formula mass and molar mass.
6. Convert mass to moles and moles to mass.
7. Calculate the number of particles in a given quantity of substance.
8. Solve combined mass–mole–particle problems using the TECHMED five-step method.

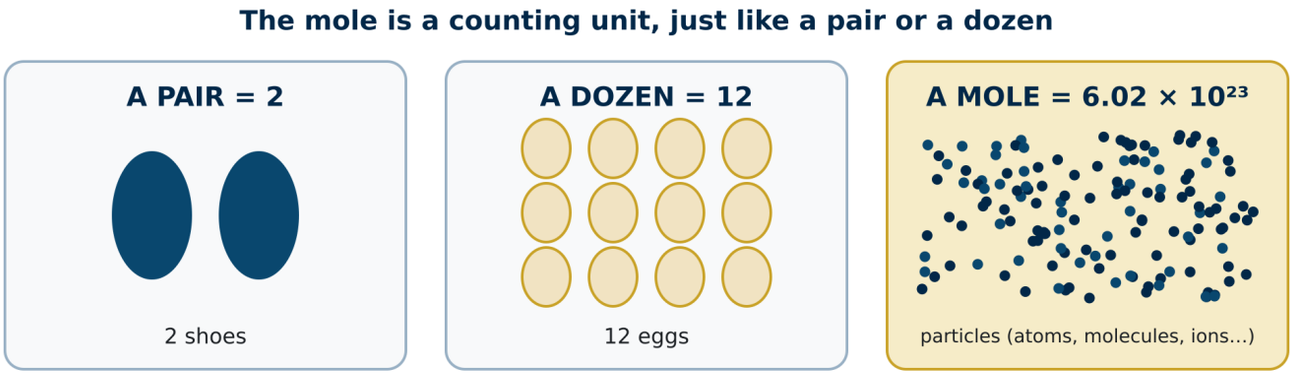
1.1 Why Do Chemists Need the Mole Concept?

Imagine someone gives you a 50 kg bag of rice and asks you to count every grain. Would you count them one after another? Of course not. You would measure the rice by **mass** instead.

Chemistry has the same problem. Atoms, molecules and ions are extremely tiny, and even a small quantity of a substance contains an enormous number of particles. Chemists keep track of them with a special counting unit called the **mole**. The mole connects the invisible world of particles to quantities we can weigh and measure in the laboratory.

> **KEY CONCEPT**
>
> The **mole** is the SI unit of **amount of substance**. It is a counting unit for a very large number of specified particles.

Counting unit	Number it represents
A pair	2 items
A dozen	12 items
A mole	about 6.02×10^{23} specified particles



Particles are far too small to count one by one, so chemists count them in moles — by weighing.

Figure 1.1 — The mole as a counting unit

TECHMED TIP

Whenever you see the word **mole**, think **counting unit**. "One mole" means a fixed, very large number of particles — just as "one dozen" means 12.

PRACTICE EXAMPLE: YOUR TURN!

1.1.1 **[WAEC]** The number of particles in one mole of a chemical compound is the

A. atomic number
B. Avogadro's number
C. mass number
D. oxidation number

1.1.2 **[TECHMED Practice Question]** Why do chemists measure substances by mass instead of counting their particles directly?

Solutions are at the end of the chapter.

1.2 What Exactly Is a Mole?

One mole contains exactly **6.02214076 × 10²³ specified elementary entities**. For SSCE and UTME calculations we use:

$$N_a = 6.02 \times 10^{23}\ mol^{-1}$$

This number is the **Avogadro constant**. The "entities" may be atoms, molecules, ions, electrons or formula units — whatever the question specifies.

- 1 mole of carbon atoms contains 6.02×10^{23} carbon atoms.
- 1 mole of water contains 6.02×10^{23} water molecules.
- 1 mole of sodium ions contains 6.02×10^{23} sodium ions.
- 1 mole of oxygen gas contains 6.02×10^{23} oxygen **molecules** (O_2).

The number stays the same; only the type of particle changes.

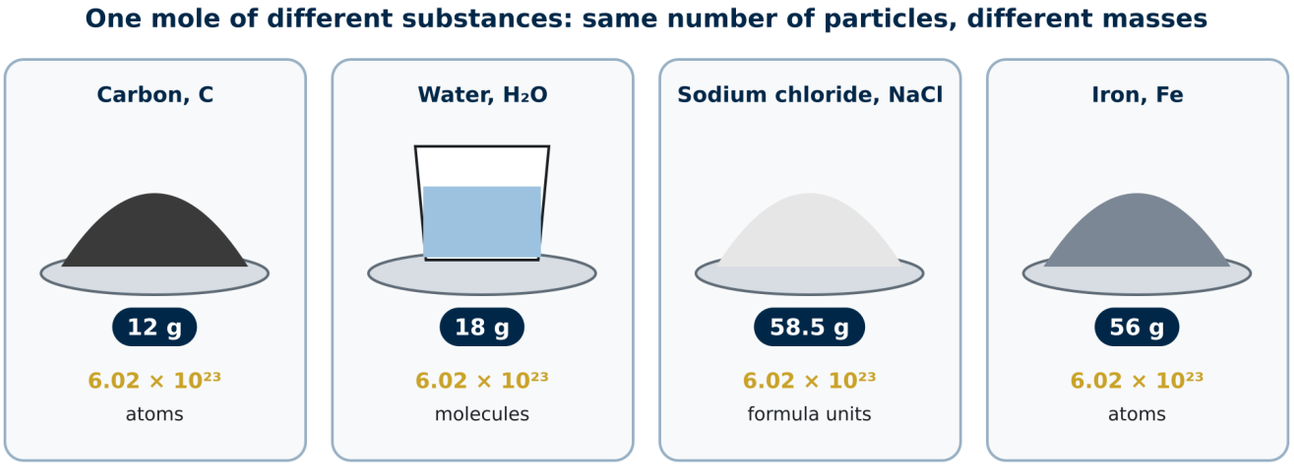


Figure 1.2 — One mole of four different substances

COMMON MISTAKE

Do not say that one mole always contains 6.02×10^{23} **atoms**. One mole contains that number of **specified entities**. Always check whether the question asks for atoms, molecules or ions.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[WAEC] Which of the following quantities is the same for one mole of $Br_2(l)$ and one mole of $He(g)$?

A. Number of molecules
B. Number of atoms
C. Total mass of the particles
D. Volume occupied at s.t.p.

Step 1: Apply the definition. One mole of any substance contains 6.02×10^{23} of its particles. Br_2 exists as molecules; helium exists as single atoms, which are its molecules.

Step 2: Check atoms. One mole of Br_2 contains 2 moles of atoms; one mole of He contains 1 mole of atoms. Not equal.

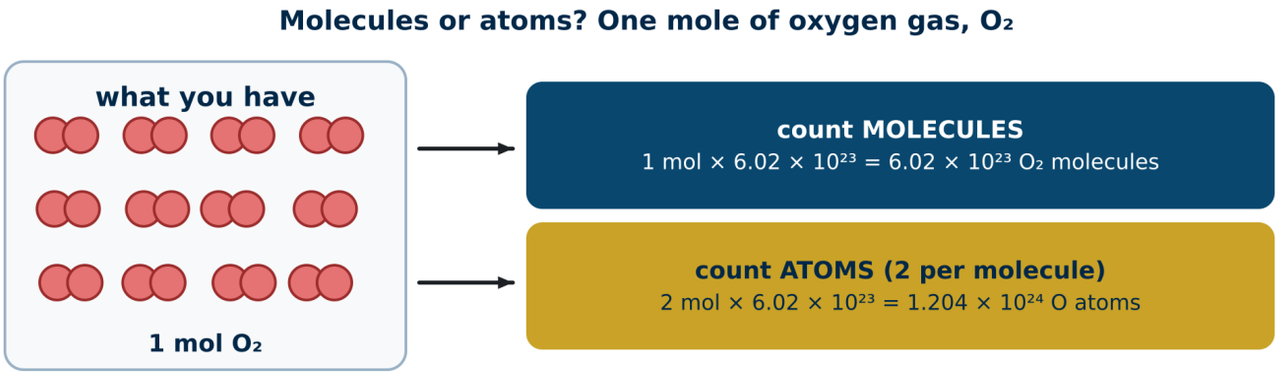
Step 3: Check mass and volume. Their molar masses differ (160 g and 4 g). Br_2 is a liquid, so it does not occupy 22.4 dm^3 at s.t.p. like a gas.

Answer: A — number of molecules.

1.3 Understanding the Avogadro Constant

The Avogadro constant, $\mathbf{N_a}$, is the number of specified entities in one mole. Its unit, mol^{-1}, means "per mole". The type of particle matters. Oxygen gas exists as O_2, and each molecule contains two atoms:

- 1 mole of O_2 contains 6.02×10^{23} oxygen **molecules**.
- 1 mole of O_2 contains **2 moles** of oxygen **atoms** = $2 \times 6.02 \times 10^{23}$ = **1.204×10^{24}** atoms.



The same applies to H_2, N_2, Cl_2, and to every atom inside a molecule (CO_2 has 3 atoms).

Figure 1.3 — One mole of O_2: molecules versus atoms

UTME TRAP ALERT

One mole of O_2 is **not** one mole of oxygen atoms. It is one mole of molecules containing two moles of atoms. The same applies to H_2, N_2, Cl_2 and every other diatomic gas.

PRACTICE EXAMPLE: YOUR TURN!

1.3.1 **[WAEC]** How many atoms are there in one mole of hydrogen gas? [Avogadro's constant = 6.02×10^{23} mol^{-1}]

A. 1.2×10^{23}
B. 6.0×10^{23}
C. 1.2×10^{24}
D. 6.0×10^{24}

1.3.2 **[TECHMED Practice Question]** How many molecules are present in one mole of carbon dioxide?

1.3.3 **[TECHMED Practice Question]** How many moles of oxygen atoms are contained in one mole of O_2?

Solutions are at the end of the chapter.

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1.4 Types of Particles: What Are We Actually Counting?

1.4.1 Atoms

Atoms are the basic units of elements. Examples: helium (He), sodium (Na), iron (Fe), carbon (C). One mole of iron contains 6.02×10^{23} iron atoms.

1.4.2 Molecules

Molecules are groups of two or more atoms chemically bonded together: O_2, H_2O, CO_2, NH_3. One mole of water contains 6.02×10^{23} water molecules, and because each molecule has 2 H atoms and 1 O atom, it also contains **2 moles of H atoms**, **1 mole of O atoms** and **3 moles of atoms in total**.

1.4.3 Ions

Ions are atoms or groups of atoms that carry a charge. **Cations** are positive (Na^+, Ca^{2+}, Al^{3+}); **anions** are negative (Cl^-, O^{2-}, SO_4^{2-}). An ion may be monatomic (Na^+) or polyatomic (SO_4^{2-}).

1.4.4 Formula units

Ionic compounds such as sodium chloride do not exist as separate molecules; their ions form a giant lattice. A **formula unit** is the simplest whole-number ratio of ions in the compound. One formula unit of $CaCl_2$ contains 1 Ca^{2+} and 2 Cl^-, so one mole of $CaCl_2$ contains 1 mole of Ca^{2+} and **2 moles** of Cl^-.

Substance	Particle being counted
Fe	Iron atoms
O_2, H_2O, CO_2	Molecules
NaCl, $CaCl_2$	Formula units
Na^+, Cl^-	Ions

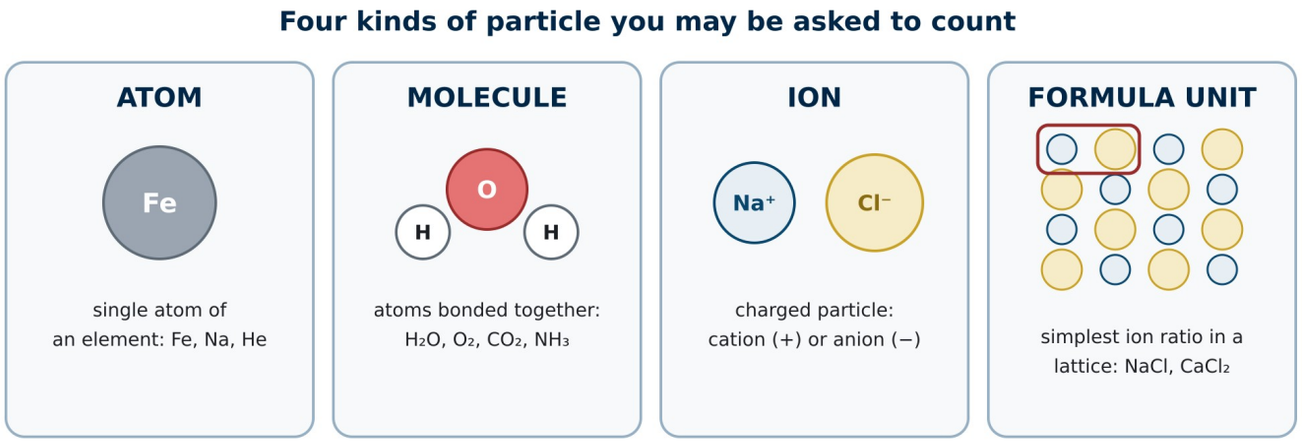


Figure 1.4 — Atoms, molecules, ions and formula units

COMMON MISTAKE

Do not assume every substance exists as separate molecules. Water consists of molecules; solid sodium chloride is an ionic lattice described by formula units.

PRACTICE EXAMPLE: YOUR TURN!

1.4.1 **[NECO]** The number of hydrogen ions in 1.0 dm^3 of a 0.25 mol dm^{-3} tetraoxosulphate(VI) acid solution is [Avogadro constant = 6.02×10^{23}]

A. 1.45×10^{23}
B. 1.50×10^{23}
C. 3.00×10^{23}
D. 3.15×10^{23}
E. 6.00×10^{23}

1.4.2 **[TECHMED Practice Question]** What type of particle is represented by Na^+?

1.4.3 **[TECHMED Practice Question]** What particle unit is used to describe solid NaCl?

1.4.4 **[TECHMED Practice Question]** How many chloride ions are represented by one formula unit of $CaCl_2$?

Solutions are at the end of the chapter.

1.5 Relative Atomic Mass (A_r)

Actual atomic masses are far too small for everyday calculations, so chemists compare them with a standard.

KEY CONCEPT

The **relative atomic mass** (A_r) of an element is the **weighted average** mass of its atoms compared with **one-twelfth of the mass of a carbon-12 atom**. It has **no unit**.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[TECHMED Practice Question] The relative atomic mass of magnesium is 24. What does this value represent?

Relative atomic mass compares the average mass of an atom with one-twelfth of the mass of a carbon-12 atom.

So A_r(Mg) = 24 means the average magnesium atom is **24 times as heavy as one-twelfth of a carbon-12 atom**.

It does **not** mean that one magnesium atom weighs 24 g.

Answer: The average mass of a magnesium atom is 24 times one-twelfth of the mass of a carbon-12 atom.

1.5.1 Why is it a weighted average? Isotopes

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons, so they have different masses. Most elements are a natural mixture of isotopes. The relative atomic mass takes the **abundance** of each isotope into account. That is why chlorine's A_r is 35.5: natural chlorine is roughly 75% ^{35}Cl and 25% ^{37}Cl.

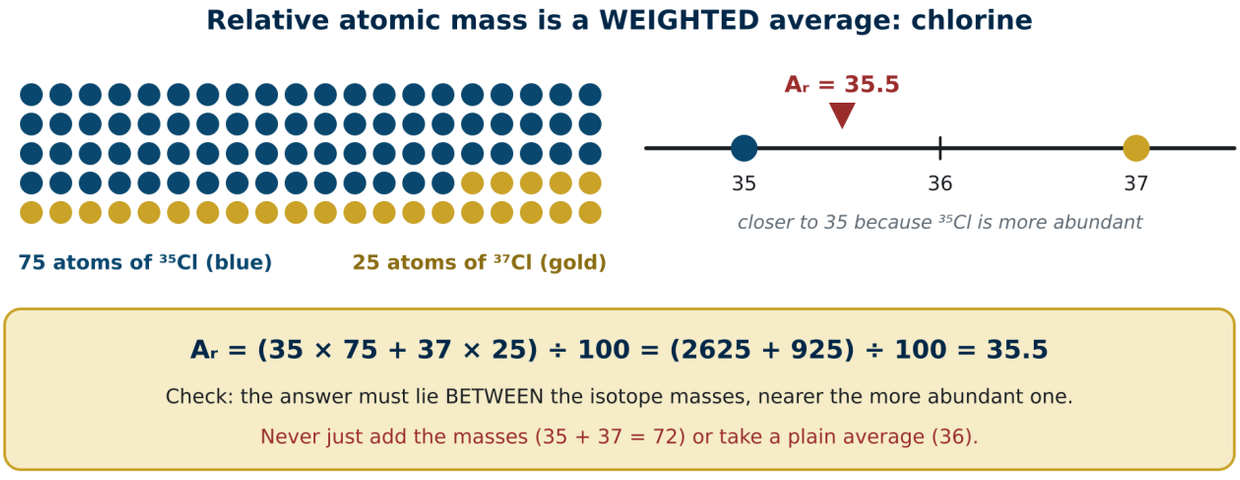


Figure 1.5 — Weighted average: the relative atomic mass of chlorine

KEY CONCEPT: Formula

A_r = (mass₁ × %₁ + mass₂ × %₂) ÷ 100

If abundances are given as a ratio instead of percentages, divide by the **sum of the ratio** instead of 100.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[WAEC] Naturally occurring boron is made up of 19.9% ^{10}B and 80.1% ^{11}B. The relative atomic mass of boron is

A. 21.0
B. 10.8
C. 10.5
D. 10.0

Given: ^{10}B = 19.9%; ^{11}B = 80.1%.

Formula: A_r = (10 × 19.9 + 11 × 80.1) ÷ 100.

Calculation: = (199 + 881.1) ÷ 100 = 1080.1 ÷ 100 = 10.80.

Check: the answer must lie between 10 and 11, closer to 11 because ^{11}B is more abundant. 10.8 fits.

Answer: B — 10.8.

UTME TRAP ALERT

Option A (21.0) is simply 10 + 11. Never add the isotope masses — **weight** them by abundance. A quick check: the average must lie between the lightest and heaviest isotope.

Element	Symbol	A_r	Element	Symbol	A_r
Hydrogen	H	1	Magnesium	Mg	24
Carbon	C	12	Aluminium	Al	27
Nitrogen	N	14	Sulphur	S	32
Oxygen	O	16	Chlorine	Cl	35.5
Sodium	Na	23	Calcium	Ca	40
Potassium	K	39	Iron	Fe	56

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PRACTICE EXAMPLE: YOUR TURN!

1.5.1 **[JAMB]** If the relative atomic mass of an element is not a whole number, it can be deduced that the element is

A. naturally radioactive
B. abundant in nature
C. a transition metal
D. an isotopic mixture

1.5.2 **[JAMB]** The relative atomic mass of naturally occurring lithium, consisting of 90% $^{7}_{3}Li$ and 10% $^{6}_{3}Li$, is

A. 6.9
B. 7.1
C. 6.2
D. 6.8

1.5.3 **[WAEC]** An element X has two isotopes, ^{65}X and ^{66}X, with relative abundances of 60% and 40% respectively. The relative atomic mass of X is

A. 66.00
B. 65.50
C. 65.00
D. 65.40

1.5.4 **[JAMB]** An element X with relative atomic mass 16.2 contains two isotopes: ^{16}X with relative abundance 90% and ^{m}X with relative abundance 10%. The value of m is

A. 14
B. 12
C. 18
D. 16

1.5.5 **[TECHMED Practice Question]** The relative atomic mass of oxygen is 16. What does this value represent?

Solutions are at the end of the chapter.

1.6 Relative Molecular Mass (M_r)

The **relative molecular mass** (M_r) is the sum of the relative atomic masses of all the atoms in one molecule. It has no unit. Multiply each A_r by the number of atoms of that element in the formula, then add.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[TECHMED Practice Question] Calculate the relative molecular mass of water, H_2O. [H = 1, O = 16]

H_2O contains 2 hydrogen atoms and 1 oxygen atom.

$M_r(H_2O) = (2 \times 1) + 16 = 2 + 16 = 18$.

Answer: 18.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[TECHMED Practice Question] Calculate the relative molecular mass of carbon dioxide, CO_2. [C = 12, O = 16]

CO_2 contains 1 carbon atom and 2 oxygen atoms.

$M_r(CO_2) = 12 + (2 \times 16) = 12 + 32 = 44$.

Answer: 44.

PRACTICE EXAMPLE: YOUR TURN!

1.6.1 **[JAMB]** Calculate the relative molecular mass of limestone, $CaCO_3$. [Ca = 40, C = 12, O = 16]

A. 56
B. 100
C. 76
D. 90

1.6.2 **[TECHMED Practice Question]** Calculate the relative molecular mass of ammonia, NH_3. [N = 14, H = 1]

1.6.3 **[TECHMED Practice Question]** Calculate the relative molecular mass of sulphur(IV) oxide, SO_2. [S = 32, O = 16]

Solutions are at the end of the chapter.

1.7 Relative Formula Mass

Ionic compounds do not exist as separate molecules, so for them we calculate the **relative formula mass**: the sum of the relative atomic masses represented in the formula. The method is identical.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[TECHMED Practice Question] Calculate the relative formula mass of sodium chloride, NaCl. [Na = 23, Cl = 35.5]

NaCl represents 1 Na^+ and 1 Cl^-.

Relative formula mass = 23 + 35.5 = 58.5.

Answer: 58.5.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[TECHMED Practice Question] Calculate the relative formula mass of aluminium tetraoxosulphate(VI), $Al_2(SO_4)_3$. [Al = 27, S = 32, O = 16]

Step 1: Count the atoms. Al = 2; S = 1 × 3 = 3; O = 4 × 3 = 12.

Step 2: Multiply. Al = 2 × 27 = 54; S = 3 × 32 = 96; O = 12 × 16 = 192.

Step 3: Add. 54 + 96 + 192 = 342.

Answer: 342.

Reading a formula with brackets: $Al_2(SO_4)_3$

$Al_2(SO_4)_3$

The subscript outside the bracket multiplies EVERY atom inside it.

Atom	How many?	Mass contribution
Al	2	2 × 27 = 54
S	1 × 3 = 3	3 × 32 = 96
O	4 × 3 = 12	12 × 16 = 192
Relative formula mass = 54 + 96 + 192 = 342		

Figure 1.6 — Counting atoms in a formula with brackets

COMMON MISTAKE

A subscript **outside a bracket multiplies every atom inside it**. In $Al_2(SO_4)_3$ there are 3 sulphur atoms and 12 oxygen atoms, not 1 and 4.

PRACTICE EXAMPLE: YOUR TURN!

1.7.1 **[WAEC]** The mass of one mole of $(NH_4)_2CO_3$ is [O = 16.0, N = 14.0, C = 12.0, H = 1.0]

A. 66.0 g
B. 76.0 g
C. 80.0 g
D. 96.0 g

1.7.2 **[TECHMED Practice Question]** Calculate the relative formula mass of calcium oxide, CaO. [Ca = 40, O = 16]

1.7.3 **[TECHMED Practice Question]** Calculate the relative formula mass of calcium trioxonitrate(V), $Ca(NO_3)_2$. [Ca = 40, N = 14, O = 16]

Solutions are at the end of the chapter.

1.8 What Is Molar Mass?

Molar mass (M) is the mass of one mole of a substance. Its unit is **g mol^{-1}**. Its numerical value equals the relative atomic, molecular or formula mass. The difference: relative mass has **no unit**; molar mass **does**.

Substance	Relative mass	Molar mass
Carbon atoms, C	12	12 g mol^{-1}
Oxygen molecules, O_2	32	32 g mol^{-1}
Water, H_2O	18	18 g mol^{-1}
Carbon(IV) oxide, CO_2	44	44 g mol^{-1}
Sodium chloride, NaCl	58.5	58.5 g mol^{-1}

Since $M(H_2O)$ = 18 g mol^{-1}: 1 mole of water weighs 18 g, 2 moles weigh 36 g, and 0.5 mole weighs 9 g.

TECHMED TIP

Do not confuse the A_r of an element with the M_r of its molecule. $A_r(O)$ = 16, but oxygen gas is O_2, so $M(O_2)$ = **32** g mol^{-1}. The same applies to H_2 (2), N_2 (28) and Cl_2 (71).

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[WAEC] If the molar mass of $X(HCO_3)_2$ is 162 g mol^{-1}, determine the relative atomic mass of X. [H = 1.0, C = 12.0, O = 16.0]

A. 40
B. 48
C. 61
D. 101

Step 1: Find the mass of one HCO_3 group. 1 + 12 + (3 × 16) = 61.

Step 2: Account for the bracket. $(HCO_3)_2$ = 2 × 61 = 122.

Step 3: Subtract. X = 162 − 122 = 40.

Trap: 61 (option C) is the mass of only one HCO_3 group; 101 comes from forgetting the bracket (162 − 61).

Answer: A — 40.

PRACTICE EXAMPLE: YOUR TURN!

1.8.1 **[JAMB]** What is the molecular mass of a monobasic alkanoic acid if 0.5 mole of the acid weighs 44 g?

A. 46 g mol^{-1}
B. 60 g mol^{-1}
C. 72 g mol^{-1}
D. 88 g mol^{-1}

1.8.2 **[NECO]** What is the molar mass of $C_6H_{12}O_6$? [H = 1, C = 12, O = 16]

A. 45
B. 61
C. 180
D. 123
E. 244

1.8.3 **[TECHMED Practice Question]** The relative molecular mass of carbon(IV) oxide is 44. What is its molar mass?

Solutions are at the end of the chapter.

1.9 Connecting Mass, Moles and Particles

Chemists measure substances by **mass**, but reactions depend on the **number of particles**. The mole connects the two.

1.9.1 Calculating moles from mass

$$n = m \div M \qquad \text{(moles = mass ÷ molar mass)}$$

n = amount of substance (mol); m = mass (g); M = molar mass (g mol^{-1}).

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[TECHMED Practice Question] Calculate the number of moles in 6.0 g of magnesium. [Mg = 24]

Given: m = 6.0 g; M(Mg) = 24 g mol^{-1}.

Formula: n = m ÷ M.

Calculation: n = 6.0 ÷ 24 = 0.25 mol.

Why it makes sense: 1 mole weighs 24 g, and 6 g is one-quarter of 24 g, so we have one-quarter of a mole.

Answer: 0.25 mol.

PRACTICE EXAMPLE: YOUR TURN!

1.9.1.1 **[WAEC]** What is the amount (in mole) of sodium trioxocarbonate(IV) in 5.3 g of the compound? (Na_2CO_3 = 106)

A. 0.05
B. 0.10
C. 0.20
D. 0.50
E. 2.00

1.9.1.2 **[NECO]** Which element contains the greatest number of moles? [He = 4, C = 12, K = 39, Zn = 65, Ag = 108]

A. 8 g of helium
B. 48 g of carbon
C. 78 g of potassium
D. 130 g of zinc
E. 216 g of silver

1.9.1.3 **[TECHMED Practice Question]** Calculate the number of moles in 12 g of carbon. [C = 12]

Solutions are at the end of the chapter.

1.9.2 Calculating the mass of a substance

m = n × M (mass = moles × molar mass)

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[TECHMED Practice Question] Calculate the mass of 2 moles of carbon(IV) oxide. [C = 12, O = 16]

Molar mass: $M(CO_2) = 12 + (2 \times 16) = 44$ g mol^{-1}.

Calculation: m = n × M = 2 × 44 = 88 g.

Answer: 88 g.

PRACTICE EXAMPLE: YOUR TURN!

1.9.2.1 **[TECHMED Practice Question]** Calculate the mass of 0.5 mole of water. [H = 1, O = 16]

1.9.2.2 **[TECHMED Practice Question]** What mass of calcium trioxocarbonate(IV), $CaCO_3$, contains 0.25 mole? [Ca = 40, C = 12, O = 16]

A. 2.5 g
B. 25 g
C. 40 g
D. 100 g

Solutions are at the end of the chapter.

1.9.3 Calculating the number of particles

$$N = n \times N_a \qquad \text{(particles = moles × Avogadro constant)}$$

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[JAMB] How many atoms are present in 6.0 g of magnesium? [Mg = 24, $N_a = 6.02 \times 10^{23}$ mol^{-1}]

A. 1.20×10^{22}
B. 2.41×10^{22}
C. 1.51×10^{23}
D. 3.02×10^{23}

Step 1: Moles. $n = m \div M = 6.0 \div 24 = 0.25$ mol.

Step 2: Particles. Magnesium exists as single atoms, so $N = 0.25 \times 6.02 \times 10^{23} = 1.505 \times 10^{23} \approx 1.51 \times 10^{23}$ atoms.

Answer: C — 1.51×10^{23}.

PRACTICE EXAMPLE: YOUR TURN!

1.9.3.1 **[WAEC]** Given that 32.0 g of sulphur contains 6.02×10^{23} sulphur atoms, how many atoms are there in 2.70 g of aluminium? [Al = 27, S = 32]

A. 6.02×10^{23}
B. 3.01×10^{22}
C. 6.02×10^{22}
D. 5.08×10^{22}

1.9.3.2 **[WAEC]** If 1 mole of sodium contains 6×10^{23} atoms, how many atoms are contained in 0.6 g of sodium? [Na = 23]

A. 1.56×10^{23}
B. 1.56×10^{22}
C. 3.6×10^{23}
D. 3.6×10^{22}

1.9.3.3 **[WAEC]** The number of atoms in 1.2 g of the carbon-12 isotope is

A. 1.00×10^{22}
B. 6.02×10^{22}
C. 1.20×10^{23}
D. 1.00×10^{23}
E. 6.02×10^{24}

Solutions are at the end of the chapter.

1.9.4 Calculating moles from the number of particles

$$n = N \div N_a$$

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[TECHMED Practice Question] How many moles are represented by 3.01×10^{23} molecules? [$N_a = 6.02 \times 10^{23}$ mol^{-1}]

$n = N \div N_a = (3.01 \times 10^{23}) \div (6.02 \times 10^{23}) = 0.5$ mol.

Answer: 0.5 mol.

PRACTICE EXAMPLE: YOUR TURN!

1.9.4.1 **[WAEC]** What is the amount of magnesium that would contain 1.20×10^{24} particles? [Mg = 24, Avogadro's constant = 6.02×10^{23}]

A. 0.5 moles
B. 2.0 moles
C. 12.0 moles
D. 24.0 moles

1.9.4.2 **[TECHMED Practice Question]** How many moles are represented by 1.204×10^{24} atoms? [$N_a = 6.02 \times 10^{23}$ mol^{-1}]

Solutions are at the end of the chapter.

1.9.5 Putting it together: the TECHMED five-step method

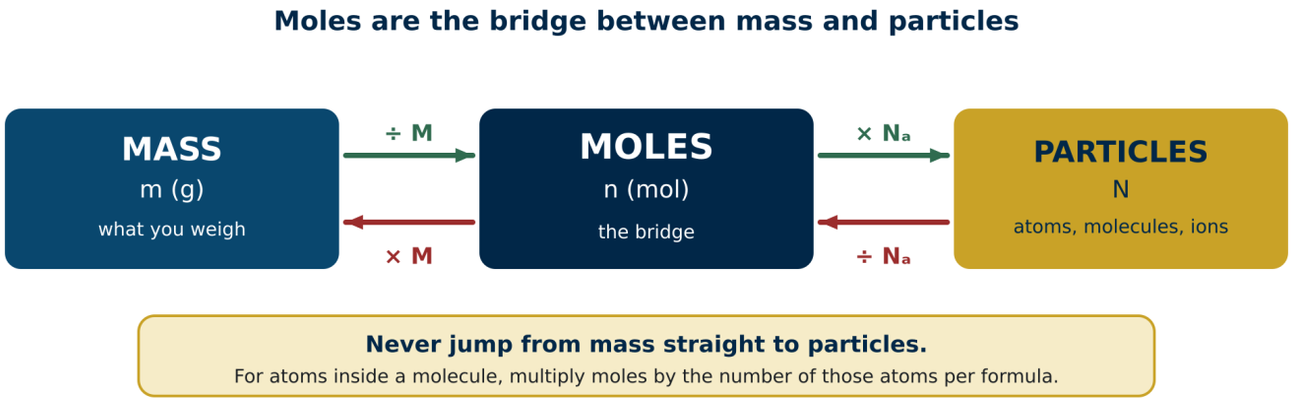


Figure 1.7 — Moles: the bridge between mass and particles

KEY CONCEPT: The TECHMED Five-Step Method

1. Given: write down what the question gives you, with units.

2. Convert to moles: $n = m \div M$ (or $n = N \div N_a$).

3. Apply the ratio: how many of the required particles are in one formula? (e.g. 3 atoms in each CO_2 molecule, 2 H^+ from each H_2SO_4.) In later chapters this becomes the mole ratio from an equation.

4. Convert to the answer: $\times N_a$ for particles, $\times$ M for mass.

5. Check: units, significant figures and the trap — atoms or molecules?

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[WAEC] The number of sulphur atoms in 3.20 g of SO_2 is [O = 16.0, S = 32.0, Avogadro constant = 6.02×10^{23}]

A. 3.01×10^{22}
B. 6.02×10^{22}
C. 6.02×10^{23}
D. 1.20×10^{24}

1. Given: m = 3.20 g of SO_2; $M(SO_2) = 32 + (2 \times 16) = 64$ g mol^{-1}.

2. Convert to moles: $n(SO_2) = 3.20 \div 64 = 0.05$ mol.

3. Apply the ratio: each SO_2 molecule contains 1 sulphur atom, so n(S) = 0.05 mol.

4. Convert: $N = 0.05 \times 6.02 \times 10^{23} = 3.01 \times 10^{22}$.

5. Check: the question asks for **sulphur atoms**, not all atoms (which would be $3 \times 3.01 \times 10^{22} = 9.03 \times 10^{22}$) and not molecules.

Answer: A — 3.01×10^{22}.

PRACTICE EXAMPLE: YOUR TURN!

1.9.5.1 **[WAEC]** The number of oxygen molecules present in 16.0 g of the gas is [O = 16, Avogadro's number = 6.02×10^{23}]

A. 6.02×10^{22}
B. 6.02×10^{23}
C. 3.01×10^{23}
D. 1.51×10^{23}

1.9.5.2 **[WAEC]** What is the mass of 6.02×10^{24} atoms of magnesium? [Mg = 24, Avogadro constant = 6.02×10^{23} atoms mol^{-1}]

A. 240 g
B. 24 g
C. 2.4 g
D. 0.24 g

1.9.5.3 **[TECHMED Practice Question]** How many molecules are present in 22 g of carbon(IV) oxide? [C = 12, O = 16, $N_a = 6.02 \times 10^{23}$ mol^{-1}]

Solutions are at the end of the chapter.

Quantity required	Formula
Moles from mass	$n = m \div M$
Mass from moles	$m = n \times M$
Particles from moles	$N = n \times N_a$
Moles from particles	$n = N \div N_a$
A_r from isotopes	$(mass_1 \times \%_1 + mass_2 \times \%_2) \div 100$

1.10 Common UTME Mistakes to Avoid

1. **Confusing mass with molar mass.** A 6 g sample of magnesium has a molar mass of 24 g mol^{-1}; they are not interchangeable.
2. **Using the wrong operation.** Moles from mass: **divide** ($n = m \div M$). Mass from moles: **multiply** ($m = n \times M$).
3. **Forgetting to work out the molar mass of a compound** before dividing.
4. **Confusing molecules with atoms.** O_2 has 2 atoms per molecule, H_2O and CO_2 have 3, NH_3 has 4.
5. **Using A_r instead of M_r for diatomic gases.** 16 g of oxygen gas is 0.5 mol of O_2, not 1 mol.
6. **Adding isotope masses** instead of weighting them by abundance.
7. **Forgetting that a bracket subscript multiplies everything inside it.**
8. **Dropping units.** Mass in g; molar mass in g mol^{-1}; amount in mol; N_a in mol^{-1}; A_r and M_r have no unit.

UTME TRAP ALERT

When asked for the number of **atoms** in a molecular substance: (1) find the number of molecules, (2) count the atoms in one molecule, (3) multiply.

1.11 Chapter 1 Quick Revision

Term	Meaning	Unit
Mole	SI unit of amount of substance; 6.02×10^{23} specified entities	mol
Avogadro constant (N_a)	Number of entities in one mole	mol^{-1}
Relative atomic mass (A_r)	Weighted average atomic mass relative to 1/12 of carbon-12	none
Relative molecular mass (M_r)	Sum of A_r of all atoms in a molecule	none
Relative formula mass	Sum of A_r in a formula unit (ionic compounds)	none
Molar mass (M)	Mass of one mole	$g\ mol^{-1}$
Isotopes	Same protons, different neutrons	—

KEY CONCEPT: Remember

1 mol O_2 = 2 mol O atoms | 1 mol H_2O = 3 mol atoms | 1 mol $CaCl_2$ = 2 mol Cl^-.

Mass ⇄ Moles ⇄ Particles: moles are always the bridge.

Your takeaway: the mole connects the number of particles in a substance to quantities that can be measured in the laboratory.

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1.12 Answers to Practice Examples

Attempt all the exercises before checking your answers.

Section 1.1

1.1.1 — B (Avogadro's number). One mole of any substance contains Avogadro's number (6.02×10^{23}) of its particles.

1.1.2 — Particles are far too small and too numerous to count. Mass can be measured easily, and the mole lets us convert mass into a number of particles.

Section 1.3

1.3.1 — C (1.2×10^{24}). Hydrogen gas is H_2. One mole contains 6.02×10^{23} molecules, each with 2 atoms: $2 \times 6.02 \times 10^{23} = 1.204 \times 10^{24} \approx 1.2 \times 10^{24}$ atoms. Option B is the number of molecules — the trap.

1.3.2 — 6.02×10^{23} carbon(IV) oxide molecules.

1.3.3 — 2 moles of oxygen atoms (each O_2 molecule has two atoms).

Section 1.4

1.4.1 — C (3.00×10^{23}). Moles of acid = 0.25 mol $dm^{-3} \times 1.0\ dm^3 = 0.25$ mol. Each H_2SO_4 releases 2 H^+, so $n(H^+) = 0.50$ mol. Number = $0.50 \times 6.02 \times 10^{23} = 3.01 \times 10^{23} \approx 3.00 \times 10^{23}$. Option B ($1.50 \times 10^{23}$) forgets the factor of 2.

1.4.2 — A sodium ion: a positively charged monatomic ion (a cation).

1.4.3 — Formula units.

1.4.4 — 2 chloride ions.

Section 1.5

1.5.1 — D (an isotopic mixture). A non-whole-number A_r is a weighted average of isotopes of different masses.

1.5.2 — A (6.9). $A_r = (7 \times 90 + 6 \times 10) \div 100 = (630 + 60) \div 100 = 6.9$.

1.5.3 — D (65.40). $A_r = (65 \times 60 + 66 \times 40) \div 100 = (3900 + 2640) \div 100 = 65.40$.

1.5.4 — C (18). $16.2 = (16 \times 90 + m \times 10) \div 100 \rightarrow 1620 = 1440 + 10m \rightarrow 10m = 180 \rightarrow m = 18$.

1.5.5 — The average mass of an oxygen atom is 16 times one-twelfth of the mass of a carbon-12 atom.

Section 1.6

1.6.1 — B (100). $40 + 12 + (3 \times 16) = 100$.

1.6.2 — $M_r(NH_3) = 14 + (3 \times 1) = 17$.

1.6.3 — $M_r(SO_2) = 32 + (2 \times 16) = 64$.

Section 1.7

1.7.1 — D (96.0 g). $(NH_4)_2 = 2 \times (14 + 4) = 36$; $CO_3 = 12 + 48 = 60$; total = 96.0 g.

1.7.2 — $40 + 16 = 56$.

1.7.3 — $40 + 2[14 + (3 \times 16)] = 40 + 2(62) = 164$.

Section 1.8

1.8.1 — D (88 g mol^{-1}). $M = m \div n = 44 \div 0.5 = 88$ g mol^{-1}. ("Monobasic alkanoic acid" is extra information here; the calculation needs only mass and moles.)

1.8.2 — C (180). $(6 \times 12) + (12 \times 1) + (6 \times 16) = 72 + 12 + 96 = 180$.

1.8.3 — 44 g mol^{-1}.

Section 1.9.1

1.9.1.1 — A (0.05). $n = 5.3 \div 106 = 0.05$ mol.

1.9.1.2 — B (48 g of carbon). Moles: He = 8 ÷ 4 = 2; C = 48 ÷ 12 = 4; K = 78 ÷ 39 = 2; Zn = 130 ÷ 65 = 2; Ag = 216 ÷ 108 = 2. Carbon has the most.

1.9.1.3 — n = 12 ÷ 12 = 1 mol.

Section 1.9.2

1.9.2.1 — $M(H_2O)$ = 18 g mol^{-1}; m = 0.5 × 18 = 9 g.

1.9.2.2 — B (25 g). $M(CaCO_3)$ = 100 g mol^{-1}; m = 0.25 × 100 = 25 g.

Section 1.9.3

1.9.3.1 — C (6.02×10^{22}). 32.0 g of S is 1 mole (6.02×10^{23} atoms). For aluminium: n = 2.70 ÷ 27 = 0.1 mol; N = 0.1 × 6.02×10^{23} = 6.02×10^{22}.

1.9.3.2 — B (1.56×10^{22}). n = 0.6 ÷ 23 = 0.0261 mol; N = 0.0261 × 6 × 10^{23} = 1.56×10^{22}.

1.9.3.3 — B (6.02×10^{22}). n = 1.2 ÷ 12 = 0.1 mol; N = 0.1 × 6.02×10^{23} = 6.02×10^{22}.

Section 1.9.4

1.9.4.1 — B (2.0 moles). n = (1.20×10^{24}) ÷ (6.02×10^{23}) = 1.99 ≈ 2.0 mol.

1.9.4.2 — n = (1.204×10^{24}) ÷ (6.02×10^{23}) = 2 mol.

Section 1.9.5

1.9.5.1 — C (3.01×10^{23}). Oxygen gas is O_2 (M = 32 g mol^{-1}). n = 16.0 ÷ 32 = 0.5 mol; N = 0.5 × 6.02×10^{23} = 3.01×10^{23} molecules. Option B comes from using 16 as the molar mass — the diatomic trap.

1.9.5.2 — A (240 g). n = (6.02×10^{24}) ÷ (6.02×10^{23}) = 10 mol; m = 10 × 24 = 240 g.

1.9.5.3 — $M(CO_2)$ = 44 g mol^{-1}; n = 22 ÷ 44 = 0.5 mol; N = 0.5 × 6.02×10^{23} = 3.01×10^{23} molecules.

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1.13 Chapter 1 Practice Questions

Attempt all questions without checking your notes. Show your working for calculations.

1. **[JAMB]** What is the molar mass of a substance if 0.4 mole of the substance has a mass of 25.0 g?

A. 62.5 g mol^{-1}
B. 40.0 g mol^{-1}
C. 6.3 g mol^{-1}
D. 2.5 g mol^{-1}

2. **[JAMB]** What is the number of moles in 44 g of carbon(IV) oxide, CO_2? [C = 12, O = 16]

A. 0.5 mole
B. 1.0 mole
C. 2.0 moles
D. 4.4 moles

3. **[JAMB]** How many molecules are present in 4.0 moles of glucose? [$N_a = 6.02 \times 10^{23}$ mol^{-1}]

A. 1.51×10^{23}
B. 2.41×10^{24}
C. 3.01×10^{23}
D. 6.02×10^{24}

4. **[WAEC]** Given that 1 mole of carbon-12 weighs 12.0 g, how many atoms are there in 24.0 g of magnesium? [C = 12, Mg = 24, $N_a = 6.02 \times 10^{23}$ mol^{-1}]

A. $\frac{1}{2} \times 6.02 \times 10^{23}$
B. $1 \times 6.02 \times 10^{23}$
C. $2 \times 6.02 \times 10^{23}$
D. $3 \times 6.02 \times 10^{23}$
E. $4 \times 6.02 \times 10^{23}$

5. **[WAEC]** The relative atomic mass of chlorine is not a whole number because

A. it is the weighted average mass of all its various isotopes
B. it is the average mass of all the isotopes of the element
C. the element has strong oxidizing ability
D. the atom contains a higher number of neutrons than protons

6. **[WAEC]** An element Q contains 69% of ^{63}Q and 31% of ^{65}Q. What is the relative atomic mass of Q?

A. 63.0
B. 63.6
C. 65.0
D. 69.0

7. **[WAEC]** An element X has isotopic masses of 6 and 7. If the relative abundance is 1 to 12.5 respectively, what is the relative atomic mass of X?

A. 6.0
B. 6.1
C. 6.9
D. 7.0

8. **[WAEC]** If 100 atoms of element X contain 70 atoms of ^{9}X and 30 atoms of ^{11}X, calculate the relative atomic mass of X.

A. 9.6
B. 10.6
C. 11.6
D. 20.0

9. **[WAEC]** A gaseous mixture contains 0.256 g of hydrogen and 2.00 g of oxygen. The mole ratio of hydrogen to oxygen in the mixture is [H = 1.00, O = 16.0]

A. 1:1
B. 2:1
C. 1:4
D. 8:1

10. **[TECHMED Practice Question]** The number of particles in one mole of oxygen gas is

A. 6.02×10^{23} oxygen atoms
B. 6.02×10^{23} oxygen molecules
C. 3.01×10^{23} oxygen atoms
D. 1.204×10^{23} oxygen molecules

11. **[TECHMED Practice Question]** Which of the following quantities has the unit g mol^{-1}?

A. Relative atomic mass
B. Relative molecular mass
C. Molar mass
D. Avogadro constant

12. **[TECHMED Practice Question]** Which of the following substances is most appropriately described using formula units rather than molecules?

A. Water
B. Oxygen gas
C. Sodium chloride
D. Carbon(IV) oxide

13. **[TECHMED Practice Question]** How many moles of calcium trioxocarbonate(IV), $CaCO_3$, are present in 2.5 g of the compound? [Ca = 40, C = 12, O = 16]

A. 0.0025 mol
B. 0.025 mol
C. 0.25 mol
D. 2.5 mol

14. **[TECHMED Practice Question]** Calculate the number of atoms present in 8.8 g of carbon(IV) oxide. [C = 12, O = 16, $N_a = 6.02 \times 10^{23}$ mol^{-1}]

A. 1.204×10^{23}
B. 3.6×10^{23}
C. 6.02×10^{23}
D. 1.806×10^{23}

15. **[TECHMED Practice Question]** How many moles of chloride ions are present in 0.5 mole of calcium chloride, $CaCl_2$?

A. 0.25 mol
B. 0.5 mol
C. 1.0 mol
D. 2.0 mol

Answer Key and Explanations

1 — A (62.5 g mol^{-1}). M = m ÷ n = 25.0 ÷ 0.4 = 62.5 g mol^{-1}.

2 — B (1.0 mole). $M(CO_2)$ = 44 g mol^{-1}; n = 44 ÷ 44 = 1.0 mol.

3 — B (2.41×10^{24}). N = $4.0 \times 6.02 \times 10^{23} = 2.408 \times 10^{24} \approx 2.41 \times 10^{24}$ molecules.

4 — B ($1 \times 6.02 \times 10^{23}$). n(Mg) = 24.0 ÷ 24 = 1 mol, so it contains $1 \times 6.02 \times 10^{23}$ atoms.

5 — A. The A_r is a **weighted** average that accounts for the abundance of each isotope. B misses the weighting.

6 — B (63.6). (63 × 69 + 65 × 31) ÷ 100 = (4347 + 2015) ÷ 100 = 63.62 ≈ 63.6.

7 — C (6.9). The abundances are a ratio, so divide by its sum: (6 × 1 + 7 × 12.5) ÷ (1 + 12.5) = 93.5 ÷ 13.5 = 6.93 ≈ 6.9.

8 — A (9.6). (9 × 70 + 11 × 30) ÷ 100 = (630 + 330) ÷ 100 = 9.6.

9 — B (2:1). Hydrogen and oxygen exist as H_2 and O_2. $n(H_2) = 0.256 \div 2 = 0.128$ mol; $n(O_2) = 2.00 \div 32 = 0.0625$ mol. Ratio = 0.128 : 0.0625 ≈ 2 : 1.

10 — B. Oxygen gas exists as O_2; one mole contains 6.02×10^{23} O_2 molecules.

11 — C. Molar mass is in g mol^{-1}. A_r and M_r have no unit; the Avogadro constant is in mol^{-1}.

12 — C. Sodium chloride is ionic (a giant lattice), so it is described by formula units.

13 — B (0.025 mol). $M(CaCO_3) = 100$ g mol^{-1}; $n = 2.5 \div 100 = 0.025$ mol.

14 — B (3.6×10^{23}). $n(CO_2) = 8.8 \div 44 = 0.2$ mol; each molecule has 3 atoms; $N = 0.2 \times 3 \times 6.02 \times 10^{23} = 3.61 \times 10^{23} \approx 3.6 \times 10^{23}$. Option A ($1.204 \times 10^{23}$) is the number of molecules.

15 — C (1.0 mol). Each $CaCl_2$ contains 2 Cl^-, so 0.5 × 2 = 1.0 mol.

CHAPTER 1 COMPLETE

You have learnt what a mole means, how to identify the particles being counted, how to calculate relative and molar masses, and how to move between mass, moles and particles. These are the foundations of quantitative chemistry. In the next chapter, you will interpret chemical formulae, calculate percentage composition, and determine empirical and molecular formulae.

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Chapter 8 — The Mole Concept Question-Pattern Library

Recognise the pattern. Choose the right relationship. Solve with confidence.

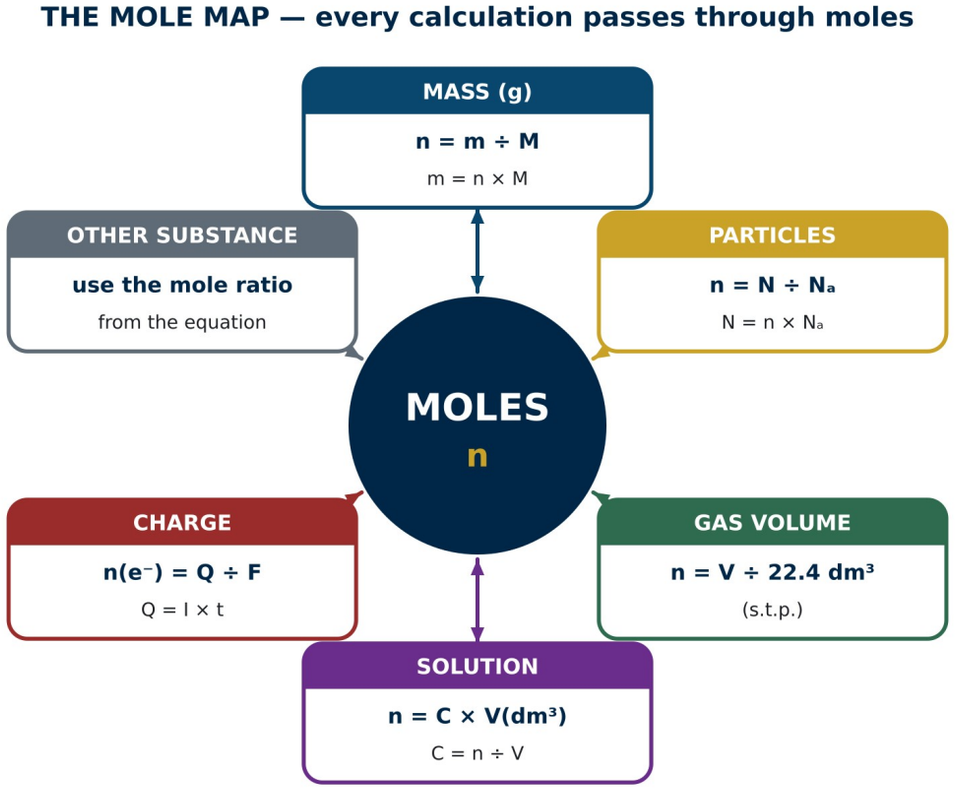
> **UTME FOCUS: How to use this chapter**
>
> Chapters 1–7 taught the ideas. This chapter trains **recognition**: seeing, within seconds, which of **19 patterns** a question belongs to and which route solves it. Each pattern has a **pattern card** (how to spot it, the route, the trap), one **worked example**, and **YOUR TURN!** questions. Answers are at the end of the chapter. Chapter 9 then gives you the full question bank for timed practice.

What You Will Learn

1. Identify what a question gives and what it requires.
2. Match a question to one of 19 calculation patterns.
3. Apply the correct route through moles for each pattern.
4. Avoid the trap examiners attach to each pattern.
5. Check every answer for size and unit.

8.1 How to Recognise a Mole-Concept Question

Almost every calculation in this book converts **what you are given** into **moles**, uses a **ratio**, and converts moles into **what is required**. The mole map shows the conversions you will use again and again.

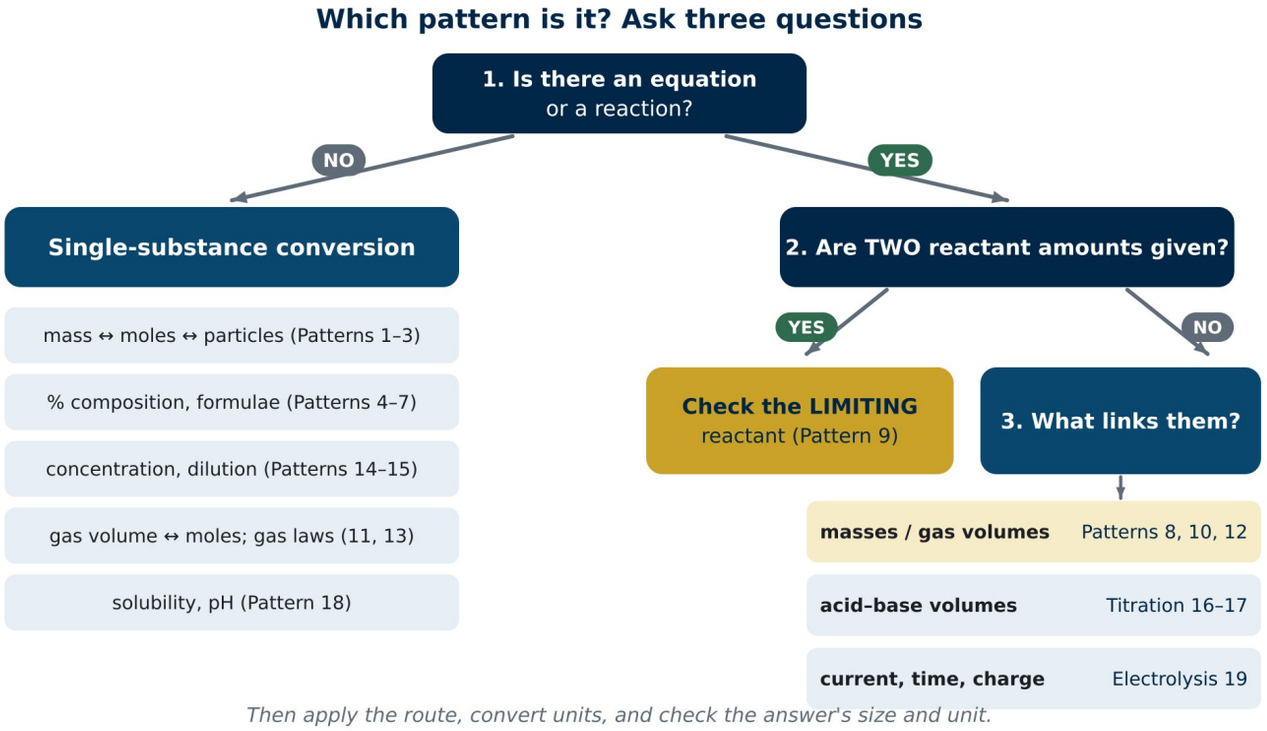


Convert what you are GIVEN into moles → use the ratio → convert moles into what is REQUIRED.

Figure 8.1 — The mole map

The question gives you…	It is probably testing…
mass and molar mass	moles (Pattern 1)

The question gives you…	It is probably testing…
moles or mass and Avogadro's constant	number of particles (Pattern 2)
mass and volume of a gas	molar mass (Pattern 3)
a formula	percentage composition (Pattern 4)
percentages or masses of elements	empirical formula (Pattern 5)
empirical formula + molar mass or V.D.	molecular formula (Pattern 6)
masses of CO_2 and H_2O from burning	combustion analysis (Pattern 7)
an equation + one quantity	reacting mass or volume (Pattern 8)
an equation + two reactant quantities	limiting reactant (Pattern 9)
a mixture and a product mass	composition of mixture (Pattern 10)
moles or mass of a gas at s.t.p.	gas volume (Pattern 11)
gas volumes in an equation	volume ratio (Pattern 12)
P, V, T changes	gas laws (Pattern 13)
mass/moles and volume of solution	concentration (Pattern 14)
water added to a solution	dilution (Pattern 15)
acid and base volumes and concentrations	titration (Pattern 16)
titration data + a sample mass	molar mass or purity (Pattern 17)
solubility or $[H^+]$	solubility and pH (Pattern 18)
current, time, charge or faradays	electrolysis (Pattern 19)



Then apply the route, convert units, and check the answer's size and unit.

Figure 8.2 — Choosing the pattern

TECHMED TIP: The TECHMED solving routine

1. Identify the given information and exactly what is required. **2. Convert** units, then convert the given quantity to moles. **3. Apply** the ratio (from the formula or balanced equation). **4. Convert** moles to the required quantity. **5. Check** the size and the unit of your answer, and whether the question asked for atoms, molecules, ions, grams, cm³ or dm³.

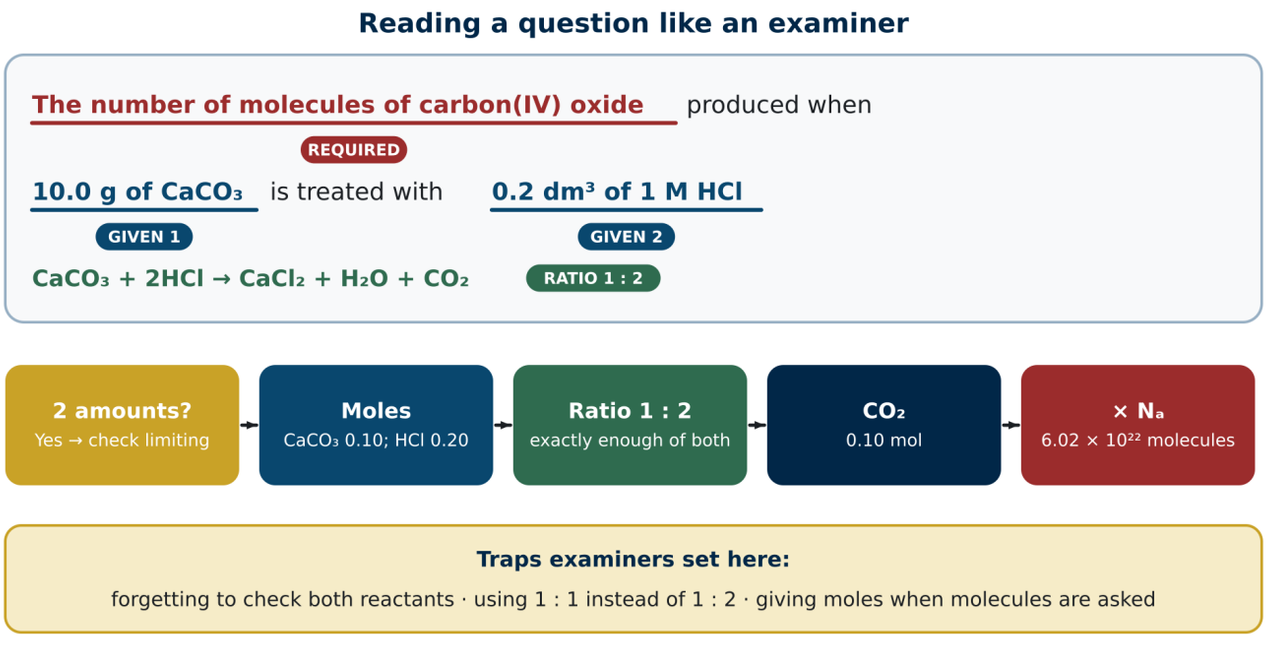


Figure 8.3 — Reading a question like an examiner

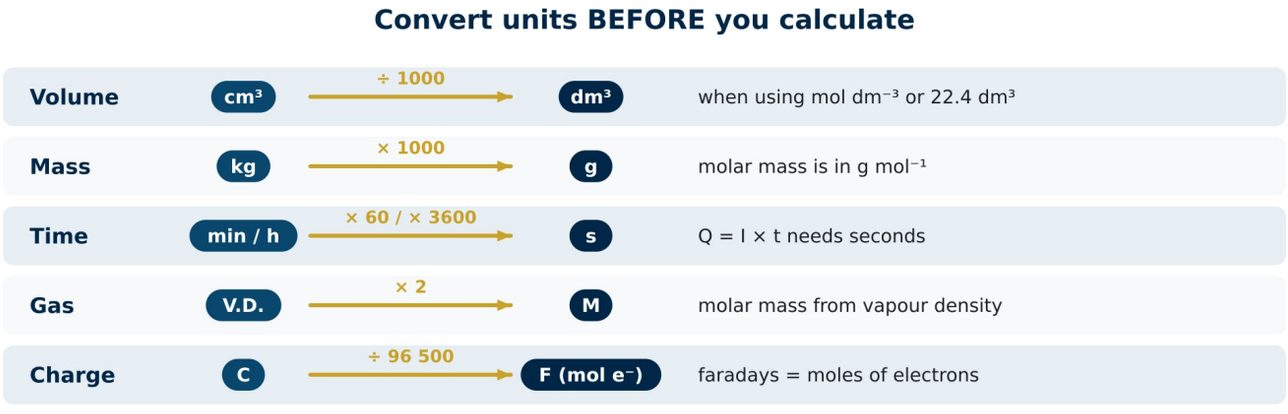


Figure 8.4 — Unit conversions to make first

PART A: Moles, Molar Mass and Particles

Pattern 1 — Mass ↔ moles

KEY CONCEPT: Pattern card

Spot it: a mass and a formula; "how many moles".

Route: n = m ÷ M; or m = n × M.

Trap: forgetting kg → g, or using the formula mass of the wrong species.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[Adapted Examination-Style Question] How many moles of calcium trioxocarbonate(IV) are there in 2.5 g of the compound? [C = 12, O = 16, Ca = 40]

A. 0.0025
B. 0.025
C. 0.25
D. 2.5

Step 1: Identify. m = 2.5 g; find n.

Step 2: Molar mass. $CaCO_3$ = 40 + 12 + 48 = 100 g mol^{-1}.

Step 3: Calculate. n = 2.5 ÷ 100 = 0.025 mol.

Step 4: Check. 2.5 g is much less than 100 g, so the answer must be much less than 1 mol.

Answer: B — 0.025.

PRACTICE EXAMPLE: YOUR TURN!

P1.1 **[Adapted Examination-Style Question]** How many moles of iron(III) oxide are contained in 1 kg of the compound? [Fe = 56, O = 16]

A. 3.125 moles
B. 0.3125 mole
C. 0.625 mole
D. 6.25 moles

Solutions are at the end of the chapter.

Pattern 2 — Moles ↔ particles

KEY CONCEPT: Pattern card

Spot it: Avogadro's constant; "number of atoms / molecules / ions".

Route: mass → moles → × N_a; multiply by atoms per formula when atoms are asked.

Trap: counting molecules when the question asks for atoms (CO_2 has 2 O atoms; P_4 has 4 P atoms).

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[Adapted Examination-Style Question] The number of atoms of oxygen present in 8.8 g of carbon(IV) oxide is [C = 12, O = 16, $N_a = 6.02 \times 10^{23}$]

A. 6.02×10^{22}
B. 2.41×10^{23}
C. 2.41×10^{22}
D. 2.41×10^{24}

Step 1: Identify. m(CO_2) = 8.8 g; required: oxygen ATOMS.

Step 2: Moles of CO_2. 8.8 ÷ 44 = 0.20 mol.

Step 3: Ratio. 1 CO_2 contains 2 O atoms → 0.40 mol O atoms.

Step 4: Particles. $0.40 \times 6.02 \times 10^{23} = 2.41 \times 10^{23}$.

Step 5: Check. A is the number of CO_2 molecules × ½; C is out by a factor of 10.

Answer: B — 2.41×10^{23}.

PRACTICE EXAMPLE: YOUR TURN!

P2.1 **[Adapted Examination-Style Question]** How many molecules of phosphorus (P_4) are present in 496 g of the substance? [P = 31, $N_a = 6.0 \times 10^{23}$]

A. 1.20×10^{23}
B. 1.20×10^{24}
C. 2.40×10^{23}
D. 2.40×10^{24}

P2.2 **[Adapted Examination-Style Question]** Which of the following contains the same number of particles as 24 g of magnesium? [Mg = 24, H = 1, O = 16, Cl = 35.5]

A. 1 g of hydrogen molecules
B. 16 g of oxygen molecules
C. 32 g of oxygen molecules
D. 35.5 g of chlorine molecules

P2.3 **[Adapted Examination-Style Question]** What mass of SO_2 contains the same number of molecules as 0.8 g of CH_4? [SO_2 = 64; CH_4 = 16]

A. 3.2 g
B. 0.32 g
C. 6.4 g
D. 0.64 g

Solutions are at the end of the chapter.

Pattern 3 — Molar mass from experimental data

KEY CONCEPT: Pattern card

Spot it: a mass and a gas volume (or "mass lost on heating").

Route: V → moles (÷ 22.4 dm^3 at s.t.p.); M = mass ÷ moles.

Trap: using cm^3 with 22.4 dm^3; forgetting that the mass of gas = mass lost.

WORKED EXAMPLE: LET'S SOLVE TOGETHER

[Adapted Examination-Style Question] On heating 1.25 g of a solid, 280 cm^3 of a gas measured at s.t.p. was evolved, leaving a residue of 0.70 g. Assuming all the mass lost was the gas, calculate the molar mass of the gas. [Molar volume = 22.4 $dm^3\ mol^{-1}$]

A. 6.4 g mol^{-1}
B. 44 g mol^{-1}
C. 32 g mol^{-1}
D. 16 g mol^{-1}

Step 1: Mass of gas. 1.25 − 0.70 = 0.55 g.

Step 2: Moles of gas. 280 cm^3 = 0.280 dm^3; 0.280 ÷ 22.4 = 0.0125 mol.

Step 3: Molar mass. 0.55 ÷ 0.0125 = 44 g mol^{-1}.

Step 4: Check. 44 = CO_2, a gas commonly lost on heating carbonates.

Answer: B — 44 g mol^{-1}.

PRACTICE EXAMPLE: YOUR TURN!

P3.1 **[JAMB]** The volume occupied by 1.58 g of a gas at s.t.p. is 500 cm^3. What is the relative molecular mass of the gas? [Molar volume = 22.4 $dm^3\ mol^{-1}$]

A. 28
B. 32
C. 44
D. 71

Solutions are at the end of the chapter.

LOCKED IN THIS PREVIEW Parts B–F (formulae, reacting quantities, gases, solutions and electrolysis), the Pattern Speed Drill and the full pattern answers continue in the full book. Continue reading on the TECHMED Library.

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Chapter 9 — The Mole Concept Question Bank

Every calculation type in one place. Practise first. Check later.

How to Use This Question Bank

This chapter brings together **102 examination-style questions** covering every calculation taught in Chapters 1–7. The questions are arranged by chapter, so you can practise a topic straight after studying it or work through the whole bank as a revision test.

1. **Attempt each question on paper first.** Write your working; do not guess.
2. **Use the TECHMED five-step method** for every calculation (below).
3. **Do not check answers question by question.** Finish a whole chapter set, then turn to the Answer Key at the end of this chapter.
4. **Mark and review.** For every question you got wrong, read the explanation and redo the calculation without looking.
5. **Take the CBT Challenge** when you are scoring 70% or more on the sets.

> **KEY CONCEPT: The TECHMED Five-Step Method**
>
> **1. Given:** write down what the question gives you, with units.
>
> **2. Convert to moles:** use mass ÷ molar mass, volume ÷ molar volume, concentration × volume, or charge ÷ 96 500.
>
> **3. Apply the ratio:** use the mole ratio from the balanced equation (or the formula).
>
> **4. Convert to the answer:** change moles into the quantity asked for — mass, volume, concentration or particles.
>
> **5. Check:** units, significant figures, and the UTME trap (cm^3 vs dm^3, atoms vs molecules, which reactant is limiting).

> **NOTE: Question labels**
>
> **[JAMB]** — a JAMB UTME past question.
>
> **[Adapted Examination-Style Question]** — a question built on the pattern of UTME, WAEC and NECO questions, or a past question with changed figures.
>
> **[TECHMED Practice Question]** — an original TECHMED question.

UTME FOCUS: The Mole Concept Cure: 100 Question Challenge (CBT)

Think you understand the Mole Concept? It's time to put that knowledge to the test.

Get ready for the **TECHMED 100 QUESTION CHALLENGE (CBT)**.

💡 100 four-option questions. ⏱️ 120 minutes under examination conditions. 🧮 On-screen calculator. 🎯 60% pass mark. 🔁 Unlimited retakes. 📚 Review correct answers and detailed explanations after submission.

This isn't just practice. It's an opportunity to diagnose your weaknesses, sharpen your calculation skills and build examination confidence.

The challenge is included in the full e-book!

🔓 Unlock *The Mole Concept Cure* in the TECHMED Resource Library:

https://techmedng.com/resources

Understand the Mole. Master the Calculations. Conquer the Questions.

TECHMED | THINK SMART SERIES.

Set A: Understanding the Mole Concept

The mole, Avogadro constant, molar mass and number of particles. Linked to Chapter 1. Questions 1–11.

1. **[JAMB]** What is the molar mass of a substance if 0.4 mole of the substance has a mass of 25.0 g?
A. 62.5 g mol^{-1}
B. 40.0 g mol^{-1}
C. 6.3 g mol^{-1}
D. 2.5 g mol^{-1}

2. **[JAMB]** Which of the following gases contains the least number of atoms?
A. 7 moles of argon
B. 4 moles of chlorine
C. 3 moles of ozone
D. 1 mole of butane

3. **[JAMB]** What is the number of moles in 44 g of carbon(IV) oxide, CO_2? [C = 12, O = 16]
A. 0.5 mole
B. 1.0 mole
C. 2.0 moles
D. 4.4 moles

4. **[JAMB]** How many molecules are present in 4.0 moles of glucose? [Avogadro constant = 6.02×10^{23} mol^{-1}]
A. 1.51×10^{23}
B. 2.41×10^{24}
C. 3.01×10^{23}
D. 6.02×10^{24}

5. **[Adapted Examination-Style Question]** A gas has a density of 1.25 g dm^{-3} at S.T.P. Calculate its molar mass. [Molar volume at S.T.P. = 22.4 dm^3 mol^{-1}]

A. 14 g mol^{-1}
B. 22.4 g mol^{-1}
C. 28 g mol^{-1}
D. 44 g mol^{-1}

6. **[Adapted Examination-Style Question]** The vapour density of a gas relative to hydrogen is 22. What is its relative molecular mass?

A. 11
B. 22
C. 44
D. 88

7. **[Adapted Examination-Style Question]** What mass of SO_2 contains the same number of molecules as 0.8 g of CH_4? [CH_4 = 16, SO_2 = 64]

A. 0.32 g
B. 3.2 g
C. 6.4 g
D. 0.64 g

8. **[Adapted Examination-Style Question]** One mole of a compound $MHCO_3$ has a mass of 84 g. Calculate the relative atomic mass of M. [H = 1, C = 12, O = 16]

A. 61
B. 42
C. 26
D. 23

9. **[Adapted Examination-Style Question]** Calculate the number of hydrogen ions present in 9.8 g of pure H_2SO_4, assuming complete ionisation. [H = 1, S = 32, O = 16; Avogadro constant = 6.02 × 10^{23} mol^{-1}]

A. 6.02 × 10^{23}
B. 6.02 × 10^{22}
C. 1.204 × 10^{23}
D. 1.204 × 10^{24}

10. **[Adapted Examination-Style Question]** $Al_2(SO_4)_3$ dissociates completely according to $Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$. How many ions are present in 0.10 mole of aluminium sulphate? [Avogadro constant = 6.02 × 10^{23} mol^{-1}]

A. 6.02 × 10^{22}
B. 1.204 × 10^{23}
C. 3.01 × 10^{23}
D. 6.02 × 10^{23}

11. **[Adapted Examination-Style Question]** A 250 cm^3 solution contains $CaCl_2$ at a concentration of 0.10 mol dm^{-3}. Assuming complete dissociation, calculate the number of chloride ions present. [Avogadro constant = 6.02 × 10^{23} mol^{-1}]

A. 1.505 × 10^{22}
B. 3.01 × 10^{22}
C. 6.02 × 10^{22}
D. 1.204 × 10^{23}

LOCKED IN THIS PREVIEW Sets B–G (formulae, stoichiometry, gases, solutions, electrolysis and organic calculations) — 90+ more timed questions — continue in the full book. Answers to Set A follow. Continue reading on the TECHMED Library.

Answer Key and Explanations

Set A: Understanding the Mole Concept

1 — A (62.5 g mol^{-1}). Molar mass = mass ÷ number of moles = 25.0 ÷ 0.4 = 62.5 g mol^{-1}.

2 — A (7 moles of argon). Argon is monoatomic, so 7 moles contain 7 moles of atoms. The other samples contain 8, 9 and 14 moles of atoms respectively.

3 — B (1.0 mole). Molar mass of CO_2 = 12 + (2 × 16) = 44 g mol^{-1}. Moles = 44 ÷ 44 = 1 mole.

4 — B (2.41 × 10^{24}). Number of molecules = moles × Avogadro constant = 4.0 × 6.02 × 10^{23} = 2.408 × 10^{24} molecules.

5 — C (28 g mol^{-1}). One mole occupies 22.4 dm^3. Its mass is 1.25 × 22.4 = 28 g. Therefore, molar mass = 28 g mol^{-1}.

6 — C (44). Relative molecular mass = 2 × vapour density = 2 × 22 = 44.

7 — B (3.2 g). Moles of CH_4 = 0.8 ÷ 16 = 0.05 mol. Equal numbers of molecules mean equal moles. Mass of SO_2 = 0.05 × 64 = 3.2 g.

8 — D (23). The mass contribution from HCO_3 is 1 + 12 + 48 = 61. Therefore, M = 84 − 61 = 23.

9 — C (1.204 × 10^{23}). Molar mass H_2SO_4 = 98. Moles = 9.8 ÷ 98 = 0.1 mol. Each mole produces 2 moles of H^+. Number of ions = 0.2 × 6.02 × 10^{23} = 1.204 × 10^{23}.

10 — C (3.01 × 10^{23}). Each formula unit produces five ions. Therefore, 0.10 mol produces 0.50 mol of ions. Number = 0.50 × 6.02 × 10^{23} = 3.01 × 10^{23} ions.

11 — B (3.01 × 10^{22}). $CaCl_2 \rightarrow Ca^{2+} + 2Cl^-$. Moles $CaCl_2$ = 0.10 × 0.250 = 0.025 mol. Chloride moles = 0.05 mol. Number = 0.05 × 6.02 × 10^{23} = 3.01 × 10^{22} ions.

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Chapter 10 — Final Revision and Examination Strategy

Your final guide to revising, calculating and performing with confidence.

UTME FOCUS: How to use this chapter

Read it **twice**: once two weeks before the examination to plan your revision, and again in the last few days as your formula sheet and checklist. It contains the **master formula sheet**, the **traps** that cost the most marks, an **examination-day method**, a **14-day revision plan**, an **error-diagnosis system**, and a 20-question **Final Readiness Check**.

10.1 The Central Principle

The mole connects mass, particles, gas volume, solution concentration and electrical charge. Almost every quantitative question follows one route:

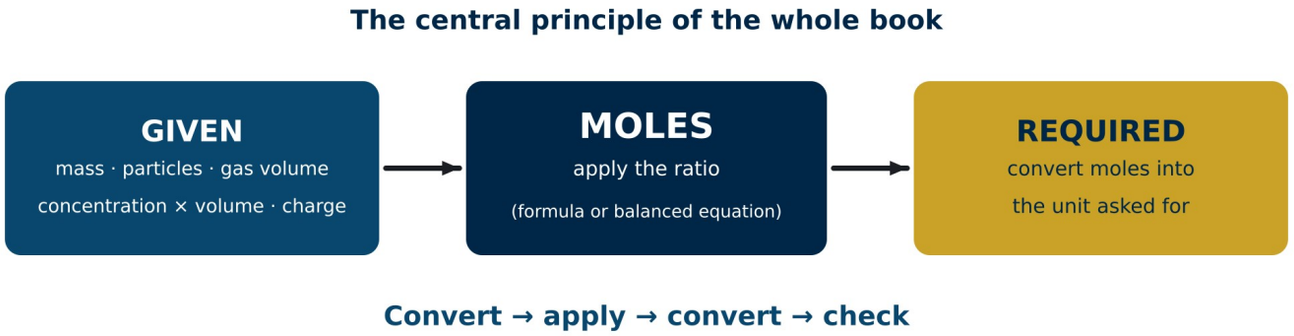


Figure 10.1 — The central principle

- **Mass:** divide by molar mass.
- **Particles:** divide by Avogadro's constant.
- **Gas volume:** divide by the molar volume given.
- **Solution:** multiply concentration by volume in dm^3.
- **Electrolysis:** Q = It, then divide by F to get moles of electrons.
- **Reaction:** use the mole ratio from the balanced equation.

10.2 The Master Formula Sheet

Area	Relationship	Remember
Mass and moles	$n = m \div M$; $m = n \times M$	M in g mol^{-1}; mass in g
Particles	$N = n \times N_a$; $N_a = 6.02 \times 10^{23}$ mol^{-1}	Atoms, molecules, ions or formula units?
% composition	% = (mass of element in 1 mol ÷ M) × 100	Include subscripts and water of crystallisation
Empirical formula	100 g → moles → ÷ smallest → whole numbers	× 2, 3, 4 to clear .5, .33, .25
Molecular formula	$n = M \div$ EF mass; $MF = (EF)_n$	M = 2 × vapour density
Combustion analysis	$n(C) = m(CO_2) \div 44$; $n(H) = 2m(H_2O) \div 18$	O by difference (C, H, O only)

Area	Relationship	Remember
Gas volume	$n = V \div 22.4$ dm³ (s.t.p.); $n = V \div 24$ dm³ (r.t.p.)	Use the molar volume stated
Gas reactions	volume ratio = mole ratio	Same temperature and pressure
Gas laws	$P_1V_1/T_1 = P_2V_2/T_2$	T in K = °C + 273; dry gas pressure = total − water vapour pressure
Concentration	$C = n \div V$; $n = C \times V$; $g\ dm^{-3} = C \times M$	V in dm³ (cm³ ÷ 1000)
Dilution	$C_1V_1 = C_2V_2$	V_2 is the total final volume
Titration	$n(A) \div a = n(B) \div b$; $(C_1V_1)/a = (C_2V_2)/b$	a, b from the balanced equation
pH	$pH = -\log[H^+]$; $[H^+][OH^-] = 10^{-14}$	H_2SO_4 gives $2H^+$ per molecule
Electrolysis	$Q = It$; $n(e^-) = Q \div F$; $m = MIt \div (zF)$	F = 96 500 C mol⁻¹; t in seconds; z from the half-equation
Percentage yield	(actual ÷ theoretical) × 100	Theoretical mass from the limiting reactant

NOTE: Constants and conversions

$N_a = 6.02 \times 10^{23}$ mol⁻¹ · F = 96 500 C mol⁻¹ · Molar volume = 22.4 dm³ (s.t.p.) or 24 dm³ (r.t.p.) · 0 °C = 273 K · 1 dm³ = 1000 cm³ · 1 kg = 1000 g · 1 h = 3600 s · Always use the values given in the question.

LOCKED IN THIS PREVIEW The traps that cost the most marks, your examination approach, the mistake-diagnosis system, the 14-day revision plan and the final readiness check continue in the full book. Continue reading on the TECHMED Library.

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TECHMED | THE MOLE CONCEPT CURE

You Have Seen How It Works

This preview gave you one full chapter and a glimpse of three more. Here is everything waiting in the full book.

In the full book	What you master
Chapter 1 — Understanding the Mole Concept	Moles, particles, A_r, M_r and molar mass (in this preview)
Chapter 2 — Chemical Formulae	Percentage composition, empirical and molecular formulae, hydrates
Chapter 3 — Equations and Stoichiometry	Reacting masses, limiting reagents, percentage yield, heat quantities
Chapter 4 — Gas Volumes	Molar volume, Avogadro's law, gas laws, ideal gas equation
Chapter 5 — Solutions and Titration	Concentration, dilution, solubility, titration and pH
Chapter 6 — Electrolysis	Faraday's laws, Q = It, masses and gas volumes at electrodes
Chapter 7 — Organic Calculations	Formulae from combustion, vapour density and general formulae
Chapter 8 — Question-Pattern Library	Every repeating mole question pattern, Parts A–F, plus a speed drill
Chapter 9 — Question Bank	102 timed questions with full explanations
Chapter 10 — Final Revision	Master formula sheet, traps, 14-day plan, readiness check

UTME FOCUS: The full book in numbers

10 chapters · **about 230 pages** · **600+ questions**, every one solved independently

300+ real JAMB, WAEC and NECO past questions, honestly labelled · **100+ worked examples** using the TECHMED five-step method

60+ custom diagrams · **wrong answer keys exposed and corrected** · appendices, glossary and progress tracker

Continue Reading on the TECHMED Library

You have already done the hardest part: you have seen that the mole concept is not magic. It is one idea — **convert to moles, use the ratio, convert back** — applied again and again. The full book takes that one idea through every calculation the UTME, WAEC and NECO can set.

1. Go to **techmedng.com/resources**.
2. Sign in or create your free TECHMED account.
3. Unlock **The Mole Concept Cure** and continue from Chapter 2 on your phone, tablet or laptop.

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UNDERSTAND THE MOLE. MASTER THE CALCULATIONS. CONQUER THE QUESTIONS.

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